Shijun Li

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Education

University of Texas at Austin, Austin, U.S. 08/2023 – Present

PhD student in Electrical and Computer Engineering, Cockrell School of Engineering

University of Science and Technology of China, Hefei, China 09/2020 – 07/2023

Master in Electronic and Information Engineering, School of Information Science and Technology

University of Science and Technology of China, Hefei, China 09/2016 – 07/2020

B.E. in Automation, School of Information Science and Technology

Preprints

- **Shijun Li**, Kaiwen Dong, Xiang Gao & Joydeep Ghosh, "Goal-Conditioned Supervised Learning for LLM Fine-Tuning", ArXiv preprint, [pdf]
- **Shijun Li**, Pranav Belligundu, Tianxin Wei, Wooseong Yang, Yu Wang & Joydeep Ghosh, "RRCM: Ranking-Driven Retrieval over Collaborative and Meta Memories for LLM Recommendation", ArXiv preprint, [pdf]
- **Shijun Li**, Yang Liu, Najmeh Sadoughi, Dongmei Wang, Xinjie Li, Zhu Liu, Rohith Mysore Vijaya Kumar, Yuzong Liu, "Retrieve What Helps, Only When It Helps: Agentic Reinforcement Learning for Selective Retrieval-Augmented Machine Translation", Preprint, [pdf]
- Hsing-Huan Chung, **Shijun Li**, Yoav Wald, Xing Han, Suchi Saria, & Joydeep Ghosh, "MILM: Large Language Models for Multimodal Irregular Time Series with Informative Sampling", ArXiv preprint, [pdf]

Publications

- **Shijun Li**, Hilaf Hasson & Joydeep Ghosh, "OMAC: A Holistic Optimization Framework for LLM-Based Multi-Agent Collaboration", International Conference on Machine Learning (**ICML 2026 Oral**), [pdf] [codes] [talk]
- **Shijun Li**, Hilaf Hasson, Jing Hu & Joydeep Ghosh, "Goal-Conditioned Supervised Learning for Multi-Objective Recommendation", Conference on Neural Information Processing Systems (**NeurIPS 2026**), [pdf] [codes] [app]
- **Shijun Li**, Yu Wang, Jin Wang, Ying Li, Joydeep Ghosh, & Anne Cocos, "LLM Reasoning for Cold-Start Item Recommendation", in Proceedings of the ACM on Web Conference 2026 (**WWW 2026**), [pdf]
- **Shijun Li**, Wenqiang Lei, Qingyun Wu, Xiangnan He, Peng Jiang & Tat-Seng Chua, "Seamlessly Unifying Attributes and Items: Conversational Recommendation for Cold-Start Users", in ACM Transactions on Information Systems **(TOIS)**, [pdf] [codes]
- Chongming Gao*, **Shijun Li***, Wenqiang Lei, Jiawei Chen, Biao Li, Peng Jiang, Xiangnan He, Jiaxin Mao & Tat-Seng Chua, "KuaiRec: A Fully-observed Dataset and Insights for Evaluating Recommender Systems", (* **Equal contribution**), International Conference on Information and Knowledge Management (**CIKM 2022**) , [pdf] [codes] [data] [presentation]
- Chongming Gao*, **Shijun Li***, Yuan Zhang, Jiawei Chen, Biao Li, Wenqiang Lei, Peng Jiang & Xiangnan He, "KuaiRand: An Unbiased Sequential Recommendation Dataset with Randomly Exposed Videos", (* **Equal contribution**), International Conference on Information and Knowledge Management (**CIKM 2022**), [pdf] [data]
- Chongming Gao, Shiqi Wang, **Shijun Li**, Jiawei Chen, Xiangnan He, Wenqiang Lei, Biao Li, Yuan Zhang & Peng Jiang, "CIRS: Bursting Filter Bubbles by Counterfactual Interactive Recommender System", in ACM Transactions on Information Systems **(TOIS)**, [pdf] [codes]
- Gangyi Zhang, Chongming Gao, Wenqiang Lei, Xiaojie Guo, **Shijun Li**, Lingfei Wu, Hongshen Chen, Zhuozhi Ding, Sulong Xu & Xiangnan He, "Vague Preference Policy Learning for Conversational Recommendation", in ACM Transactions on Information Systems **(TOIS)**, [pdf]

Experience

Amazon - Applied Science Intern; Seattle, U.S. 06/2026 – 08/2026

- Proposed and developed an agentic RL framework that learns adaptive retrieval, reasoning, and translation strategies in a multi-round manner to improve downstream translation tasks on Amazon Prime Video
- Developed joint Plackett-Luce and task-utility contrastive optimization for retriever training. Conducted experiments on various translation tasks with offline and online retrieval, which demonstrate consistent improvement

Netflix - Machine Learning Research Intern; Los Gatos, U.S. 06/2025 – 08/2025

- Developed LLM-based reasoning methodologies for recommendations in cold-start scenarios, leveraging supervised and RL-based fine-tuning approaches to further enhance reasoning capabilities
- Conducted experiments on real-world data at Netflix. Results demonstrated the superior performance of reasoning strategies on certain key metrics after fine-tuning against Netflix's base ranking model

Intuit - Data Science Intern; Mountain View, U.S. 05/2024 – 08/2024

- Applied goal-conditioned supervised learning to Intuit's recommender library, targeting to achieve better long-term rewards and reduce noise caused by users' random choices in the training data
- Conducted offline experiments on real-world dataset from Intuit's recommendation system. Results demonstrated significant improvements over the existing baseline model with minimal increase in complexity

Kuaishou Technology - Algorithm Intern; Beijing, China 03/2020 – 05/2023

- Studied the cold-start scenario and trustworthy evaluation of recommendation systems
- Implemented offline RL in real-world short video recommendation system, serving millions of people and achieving significant online improvement

University of Florida, SmartData Lab - Research Intern; Gainesville, U.S. 07/2019 – 08/2019

- Studied the growth and boundaries of grains in microstructure in an RL framework
- Encoded microstructure pictures into low-dimension expression, while denoising for the vagueness and noises

Selected Projects & Research

Project: General Optimization Framework for Multi-Agent Collaboration 07/2024 - 06/2025

Advisor: Dr. Hilaf Hasson (Intuit), Prof. Joydeep Ghosh (UT Austin)

- Developed a general framework to optimize the functionality and structure of multi-agent collaboration
- Developed evolution algorithms for general single-dimension and multi-dimension optimization

Project: Goal-conditioned Supervised Learning for Multi-Objective Learning 11/2023 - 05/2024

Advisor: Dr. Hilaf Hasson (Intuit), Dr. Jing Hu (Intuit), Prof. Joydeep Ghosh (UT Austin)

- Introduced a general supervised framework for multi-objective recommendations that selectively leverages training data with desirable long-term rewards and avoids complex architectures or optimization constraints
- Introduced a goal-choosing algorithm that can model the distribution of achievable goals over interaction sequences and choose desirable "high" goals for inference

Project: Cold-Start Recommendation by Conversational Recommendation 03/2020 - 12/2020

Advisor: Prof. Xiangnan He (USTC), Prof. Qingyun Wu (PSU), Prof. Tat-Seng Chua (NUS)

- Actively asking users' preferences by conversations helps to efficiently capture the interest of cold-start users
- Proposed a holistic framework to seamlessly solve all conversation policy questions in an end-to-end manner
- Applied Thompson Sampling to conversational recommendation for keeping EE balance in cold-start scenario

Project: Trustworthy Evaluation for Recommender Systems 03/2021 - 12/2021

Advisor: Prof. Xiangnan He (USTC), Prof. Wenqiang Lei (SCU), Dr. Peng Jiang (Kuaishou Inc.)

- Collected a fully-observed dataset on Kuaishou to support trustworthy evaluation of recommender systems
- Studied the effect of exposure rates, various biases, and data completion for recommendation system evaluation

Project: Reinfocement Learning Implementation in Real-World Recommender System 03/2022 - 07/2022

Advisor: Prof. Xiangnan He (USTC), Dr. Yuan Zhang (Kuaishou Inc.), Dr. Peng Jiang (Kuaishou Inc.)

- Designed an actor-critic based RL framework for online recommendation for millions of users on Kuaishou
- Trained the model in an offline RL manner by building and interacting with the world model
- Deployment on Kuaishou's online recommendation significantly improved users' total watch time and diversity

Professional Services & Awards

- **PC Member** for ICLR (2025-2027), KDD (2022-2024), ECML-PKDD (2024), WSDM (2022), ICLR 2027 Workshop Proposal, and the 1st Workshop on Recommendation with Generative Models on CIKM (2023)
- **Reviewer** for ACM TOIS, ACM TWEB, ACM TORS, ICML (2025-2026), NeurIPS (2024-2026), SDM 2026, and WWW (2022)
- UT Austin Engineering Fellowship, UT Austin, U.S. 2023 & 2024 & 2025 & 2026
- Illinois Distinguished Fellowship, UIUC, U.S. (declined) 2023
- Outstanding Graduate Scholarship, USTC, China 2023
- National Scholarship, Ministry of Education of China, China (for top 2% students) 2022